TP – Automated installation of a CentOS cluster

ING3 - ICC – Cloud Infrastructures

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Introduction

As cloud administrators, you will frequently face situations in which you will be required to set up an Infrastructure-as-a-Service (IaaS) platform. The first step to do so is to take a bunch of physical machines and configure a cluster, both the hardware (storage, networks, wiring, etc.) and the software (OS installation and configuration, user accounts, etc.). Let us set aside the hardware part and suppose that this has already been done for us, so that we can concentrate on the software side. Say that we have a 128-machine cluster. For obvious reasons, we will not install and configure manually an OS on each machine. Fortunately, there exist several technologies to perform an automated installation of a cluster, in which nodes boot with an empty hard drive, get an IP from the network and start downloading and installing an OS.

One of those technologies, aimed at Red Hat-based systems, is **Kickstart**.

Hands on

Using your recently-acquired expertise in KVM, you are required to set up a virtual cluster. First, you will configure a VM to work as a Kickstart server. Then, you should be able to boot any other VM that, simulating a node in the cluster, will connect to the Kickstart server at boot time and will install and configure a CentOS 7 distribution.

The following are the features (required, but not limited to) to be provided by the Kickstart server and/or the cluster nodes:

1. The Server will display two network interfaces: one to connect to the internet in NAT-mode through the host (your laptop), and a second one in a private network, exclusively devoted to the cluster. ☐

2. Any cluster node, when booting, will report its presence in the network. The Server will respond and provide it with a dynamic IP in the cluster address range. Cluster nodes will display a single network interface. ☐

3 The Server will provide the booting nodes with a way to access, download and use:

(a) a CentOS 7 distribution to install, and

(b) a Kickstart configuration file to specify the installing and post-installing configuration parameters.

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4 The CentOS 7 distribution will actually reside in the host machine (your laptop), so the Server must be able to make it available to the cluster nodes in a transparent way. ☐

5 Every cluster nodes will be installed with the following features:

1. A disk with a main and a swap partitions.
2. A password-enabled user account.
3. The cluster administrator's SSH keys and a ssh-based, passwordless configuration for the root user.

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Notes

1. You are not required to perform the aforementioned steps in order, as long as you provide a solution with the requested functionalities.
2. You can use `virsh` and/or `virt-manager` to create and handle your VMs and virtual networks.
3. Take into account that the Server must run a DHCP service in order to provide its clients with an IP address. This means that any communication between the Server and the clients must be done through a private virtual net. Make sure that your Server's DHCP service **is not accessible from outside the host** (your laptop). Otherwise it will conflict with the CY-Tech main DHCP server.

Resources

You must use the CentOS 7 isos, whose images are available at
https://buildlogs.centos.org/rolling/7/isos/x86_64/

It is recommended to use the CentOS-7-x86_64-DVD-1511.iso image for the Kickstart server and the CentOS-7-x86_64-Minimal-1511.iso for the clients.

References

1. Red Hat Enterprise Linux 7 Installation Guide: https://access.redhat.com/documentation/en-us/red_hat_enterprise_linux/7/html/installation_guide/index